

SHEIKH AFAAN FAROOQ

PhD Candidate (Computer Science) | Assistant Professor | Artificial Intelligence Researcher

Malaysian Spouse Dependent Pass Holder | Ready to Relocate

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Professional Links: Scopus | Google Scholar | LinkedIn | ResearchGate | HackerEarth | Github

PROFESSIONAL SUMMARY

PhD Candidate with **4 years** of university level teaching experience and **31 scholarly publications**. Research focuses on Artificial Intelligence, Machine Learning and Deep Learning with experience in building end-to-end classification pipelines for disorder analysis and detection across text and image data, addressing a diverse range of human and plant disorders. Also, has experience as a **Subject Matter Expert in Computer Science**, **Software Development Engineer in Test (SDET)**, knowledge of programming, Selenium automation testing, software testing methodologies and quality assurance. Widely experienced in the accreditation of programs by **ABET**, **outcome-based education**, **Academic Course Coordination and Management, MOOCs coordination** and **supervision of student projects**. Looking for an opportunity to work within a professional environment that is dynamic and offers challenges and rewards for the use of my knowledge, technical skills, research capabilities and experiences to support the success of the organization and continuously improve my skills, capabilities and career path.

CORE TECHNICAL COMPETENCIES

Artificial Intelligence	Deep Learning · Machine Learning · Feature Selection · Feature Extraction · Binary and Multi-Class Classification
Application Domains	Medical image analysis · Computer Vision · Disorder Classification · Health Informatics
Technical Skills	Python · C · C++ · Java · PHP · SQL · Java Selenium · Data Structures and Algorithms · MS Excel · Databases · Computer Organization · Software Engineering · Microcontrollers
Networks & Security	Computer Networks · Wireless Mobile Communications · Cybersecurity Fundamentals ((ISC)^2, IBM)
Academic & Quality Assurance	ABET Accreditation · Outcome-Based Education · Curriculum Delivery & Assessment · MOOC Coordination · Student Project Supervision · Research Mentoring · Social Internship Coordination
Research Practice	Scientific Writing · Independent & First-Author Research · Reviewer · Best Paper IEEE ICDSCNC-2024 · Patent Filing

ACADEMIC & PROFESSIONAL EXPERIENCE

Assistant Professor, Computer Science | Chandigarh University, Mohali, India **Nov 2022 – Present**

- **Teaching:** Delivering undergraduate and graduate Computer Science courses, including Data Structures, Artificial Intelligence, Blockchain, Data Science, Python Programming, Java Programming, Computer Architecture and Organization, and Competitive Coding, through engaging lectures and hands-on practical sessions, while designing assessments, evaluating student performance, and providing individual academic mentoring and guidance.
- **Accreditation & Quality:** Supported ABET accreditation and academic quality-assurance processes, contributing to outcome-based assessment and continuous improvement.
- **MOOCs & Experiential Learning:** Currently serving as the MOOCs Coordinator and coordinating the Social Internship course for undergraduate and graduate students, overseeing academic coordination and related activities for more than 5,000 students.
- **Project Supervision:** Worked as Project Coordinator, guiding student project teams through planning, development and academic evaluation.
- **Research:** Sustaining an active AI/ML research agenda alongside teaching, producing peer-reviewed Scopus-indexed and IEEE publications, book chapters and patents.

Subject Matter Expert, Computer Science | Vaidik Eduservices Pvt. Ltd. **Jun 2021 – Dec 2022**

- Resolved technical queries across programming, data structures and core Computer Science topics, delivering accurate, concept-focused academic solutions to learners.

- Applied strong programming and problem-solving foundations to explain algorithms and code-level solutions clearly.

Software Development Engineer in Test | Affle Pvt. Ltd. **Oct 2019 - Feb 2020**

- Automated software testing using Java Selenium to validate application functionality.
- Performed manual testing, including functional and regression testing, to identify defects and ensure software quality.

EDUCATION

PhD in Computer Science (Pursuing) | Chandigarh University, Mohali, India | Grade: A **2024 – Present**

Research focus: Machine learning and deep learning for disease detection and medical image diagnosis.

Thesis: Lung disease prediction through feature extraction using an enhanced deep learning model

Highlights: Achieved the distinction of being the first scholar in my batch to complete the PhD synopsis, demonstrating research commitment, academic dedication, and effective progress in doctoral research.

M.Tech in Information Technology | Central University of Kashmir, Ganderbal, India | Grade: A **2020 – 2022**

Thesis: Post-traumatic stress disorder (PTSD) prediction in Covid-19 survivors - A Clinical Questionnaire and Machine Learning Based Approach

Awards: Secured 2nd position in an inter-university coding competition, demonstrating strong programming, analytical thinking, problem-solving, and competitive coding skills.

B.Tech in Information Technology | National Institute of Technology (NIT), Srinagar, India | Grade: A **2015 – 2019**

Project: Depression Prediction and Analysis - A Machine Learning Approach

Highlights: Recognized for data collection from local clinics and hospitals, poem featured in yearly publication.

Schooling | Tyndale Biscoe School, Srinagar, India | Grade: A+ **2001 – 2013**

RESEARCH OUTPUT

31 total publications | 2 Scopus-indexed journal articles | 26 IEEE conference publications | 2 book chapters | 1 patent

Selected Publications

- Optimizing Medical Image Diagnosis: The Role of Feature Selection in Deep Learning Models. *Sakarya University Journal of Computer and Information Sciences*. Q3, Scopus-indexed | First author.
- Prediction of Lung Diseases Using Deep Learning Techniques: A Review. *Pertanika Journal of Science & Technology*. Q3, Scopus, ESCI | First author.
- The Deep Learning Based Analysis and Prediction of Apple Scab Disease in Apple Plants using ResNet-50 Hybrid Algorithm. *IEEE Conference Publication, IEEE Xplore*. Q4, Scopus | Single author.
- The Multi-Class Detection of Five Stages of Hepatitis C Using the Machine Learning Based Random Forest Algorithm. *IEEE Conference Publication, IEEE Xplore*. Q4, Scopus | Single author.
- AI based mathematical equation solving device | Indian Patent Registration 441463-001

SELECTED RESEARCH & TECHNICAL PROJECTS

AI-Based Disease Detection Systems | Research Projects **2019 – Present**

- Designed and implemented machine learning and deep learning classifiers for binary and multi-class disease detection from textual and image data.
- Systems include depression, PTSD, hepatitis C (five-stage classification using Machine Learning), lung-disorder (Feature Extraction – Feature Selection – Classification hybridization) and apple scab detection.
- Investigated the impact of feature selection, feature extraction and ensemble learning on deep learning accuracy in medical image diagnosis.
- Collected and organized healthcare data from hospitals to support the development, training, and evaluation of machine learning and deep learning models for disease prediction and clinical research.

Network Simulator with Full Protocol Stack | Academic Project, C++ **2017**

- Designed and implemented a network simulator in C++ modelling the complete protocol stack.

hidpre.info, Anonymous Information-Sharing Platform | Web Project **2018**

- Developed a full web application enabling anonymous information sharing.

zakatcalculator.in, Zakat Calculator | Web Project **2023**

- Developed a web-based application for calculating Zakat based on user's financial information and applicable Zakat criteria.

CERTIFICATIONS & TRAINING

- Certified in Cybersecurity, $(ISC)^2$, 2023
- Introduction to Cybersecurity, IBM, 2023
- Java, PHP and MySQL, Indian Institute of Technology (IIT) Bombay, 2017
- Ethics and Gender Equality, LinkedIn Learning, 2023
- CyberPeace First Responders and Myth Busters, 2025
- Research Methodology and Effective Teaching Learning Process, 2025
- Introduction to Databases, 2024
- Microcontroller, Arduino platform and C programming, 2024
- Basic Civil Defence Course, Regional Civil Defence Control Centre, Srinagar, 2012
- Disaster Management, Tyndale Biscoe School, Srinagar, 2006
- Stanford Study Performance Report - Critical thinking test: Qualified and Relational reasoning test: Qualified, April 2018
- Inspire Science Camp, National Institute of Technology (NIT) Srinagar, 2012
- Football Training, Tyndale Biscoe School, Srinagar, 2004

INTERNSHIPS, LEADERSHIP & SERVICE

- **PHP Intern,** Aptech Ltd. (Dec – Feb 2018)
- **Wireless Mobile Communications Training,** BSNL (Jul – Aug 2017)
- **Community Leader,** MentorBoxx (May – Jun 2021)
- **COVID-19 Response Volunteer & Fundraiser,** Caregiver Saathi, Muskurahat Foundation and Womenite (Jan – June 2021)

EXTRA CURRICULAR ACTIVITIES

- Member in football team at school level 2004 - 2011
- Member in football team at college level 2011 - 2013
- Member in football team at university level 2015 - 2016
- Member in climbing/trekking the Mahadev Peak, Tatakooti Peak, Kolahoi Peak, Harmukh Peak and the twin peaks in the Zabarwan range in 2007 - 2011

LANGUAGES

English | Urdu | Hindi | Kashmiri